PRODUCTS & COMPATIBILITY

PC Components Compatibility

Building or upgrading a gaming PC requires careful consideration of hardware compatibility. Computer components may use different physical connectors, electrical standards, and technical specifications. Even when two products are individually functional, they may not work together in the same computer.

PixelForge recommends checking compatibility before purchasing components, particularly when upgrading an existing system or assembling a computer from individual parts.

Processor and Motherboard Compatibility

The processor and motherboard must use compatible CPU sockets. For example, AMD Ryzen processors designed for the AM5 socket require a compatible AM5 motherboard, while Intel processors designed for the LGA1700 socket require a motherboard supporting that socket.

A matching socket does not always guarantee compatibility. Motherboards also require appropriate chipset support and a BIOS version that recognizes the processor.

Customers should consult the motherboard manufacturer's CPU compatibility list before purchasing a processor. This list normally identifies supported processor models and any minimum BIOS requirements.

Installing an incompatible processor can prevent the computer from starting. Customers should never force a processor into a socket or attempt to modify physical connectors.

Memory Compatibility

Modern desktop computers commonly use DDR4 or DDR5 memory. These standards are not interchangeable, and a motherboard designed for DDR4 cannot use DDR5 modules unless it has specifically designed compatible slots, which ordinary consumer motherboards do not provide.

Memory compatibility also depends on supported capacities, speeds, module configurations, and motherboard firmware.

A memory kit advertised at a particular speed may require enabling an XMP or EXPO profile in the BIOS. The maximum stable speed depends on the processor, motherboard, and memory configuration.

PixelForge recommends purchasing matched memory kits rather than combining modules with different specifications. Mixing memory may work in some systems but can cause reduced performance or stability problems.

Graphics Card Compatibility

Most modern dedicated graphics cards use a PCI Express x16 expansion slot. PCI Express generations are generally designed with backward compatibility, although available bandwidth may differ depending on the motherboard and graphics card.

Physical dimensions are an important consideration. High-performance graphics cards may occupy several expansion slots and require significant clearance inside the computer case.

Customers should compare the graphics card's length, thickness, and height with the maximum dimensions supported by the case. Additional space may be required for power connectors and cable routing.

A graphics card may also require one or more dedicated power connectors. Customers must verify that their power supply provides the correct connectors and sufficient capacity.

Power Supply Requirements

The power supply must provide enough electrical power for the complete system, including the processor, graphics card, storage devices, cooling equipment, and connected accessories.

A power supply's advertised wattage should not be the only factor considered. Build quality, efficiency, connector availability, and manufacturer recommendations are also important.

For example, a gaming PC with a high-performance graphics card may require a substantially larger power supply than a system using integrated graphics.

Customers should follow the graphics card manufacturer's recommended system power requirements and allow reasonable capacity for other components and future upgrades.

Modular power supply cables are not necessarily interchangeable between different brands or models. Using an incompatible modular cable can damage connected hardware.

Storage Compatibility

Modern motherboards may support SATA storage devices and M.2 solid-state drives. However, not all M.2 slots support the same interfaces.

An M.2 drive may use SATA or PCI Express NVMe technology, and customers must verify that the motherboard slot supports the intended drive type.

NVMe drives are available in different PCI Express generations and physical lengths. A motherboard may support a newer drive at a lower speed, depending on its capabilities.

Some motherboards share bandwidth between storage connectors and expansion slots. Installing a particular M.2 drive may disable another connector or reduce available PCI Express lanes.

Customers should consult the motherboard manual for these limitations.

Cooling and Case Compatibility

CPU coolers must support the processor socket and fit within the computer case. Large air coolers may interfere with tall memory modules or exceed the case's maximum supported cooler height.

Liquid cooling radiators require suitable mounting locations and sufficient clearance for fans, tubing, and nearby components.

Case airflow should also be considered. High-performance hardware produces heat that must be removed efficiently to maintain stable operation.

BIOS Updates and Firmware

Some compatible components require a BIOS update before they can function correctly. This is particularly relevant when installing a newer processor on an older motherboard.

BIOS updates should be performed according to the motherboard manufacturer's instructions. Interrupting a firmware update may prevent the motherboard from starting.

Customers who are uncertain about compatibility should contact PixelForge support before purchasing or installing components.

Compatibility Assistance

PixelForge can provide general guidance about component specifications and common compatibility requirements. Customers should provide the exact model numbers of their existing components when requesting assistance.

For information about complete computer configurations, see Custom PC Builds. For details about interpreting product specifications, consult Product Specifications.